DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #05B: NON-WEAPON EQUIPMENT & PROCEDURAL SYSTEM (PART 2 — PERSONAL JEWELRY & WEARABLES: `RING`, `NECKLACE`/`PENDANT`, `BRACELET`, `CHARM`/`ORNAMENT`, `SHOES`, SPATIAL POUCHES & 4 LEGENDARY ACCESSORIES)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Master Roadmap (`05A–05G`)**:
  - `05A`: Group 1 — Body Armor & Protective Attire (`117` catalog items)
  - **`05B` (This Document)**: Group 2 — **Personal Jewelry & Wearables (`RING`, `NECKLACE`/`PENDANT`, `BRACELET`, `CHARM`/`ORNAMENT`, `SHOES`, Spatial Bags, & All 4 Legendary Accessories `LEG_ACC_*`)**
  - `05C`: Group 3 — Wearable Talismans, Portable Formation Cores & Profession/Exploration Tools
  - `05D`: Group 4 — Multi-Piece Set & Synergy Architecture (`62` Resonance Sets, Paired Items, Cursed/Soul-Bound, Catalysts & Trinity)
  - `05E`: Part 5 — Holistic Overview of Non-Weapon Equipment (`216` Accessories + `117` Armors)
  - `05F`: Part 6 — Penultimate Deep Audit: Procedural Equipment Engine & Templates
  - `05G`: Part 7 — Grand Finale: Master End-to-End Audit of the Entire Equipment System

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **40-Script Load Order & Closed-Loop Supply Conservation**: Spatial pouches (`EFX_SUPPLY_EFFICIENCY`) may reduce expedition ration consumption (`15%..22%`), never mint free `meal` items out of thin air.

3. **100% Emergent Role-Driven Accessory Selection**: Assassins should naturally favor `RING`s, tanks/monks `NECKLACE`s, scholars/casters `CHARM`s, scouts/couriers `SHOES`, and caravan leaders Spatial Pouches—and pure Healers/Supports must be able to trigger their Legendary Accessories without being forced to land offensive melee hits!

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1. [ACC-LEG-01] Live Runtime Bugs Across ALL FOUR Tier-7 Legendary Accessories (`js/visitors_clues.js:598–717, 905–957`)!

Every single one of the **4 Legendary Accessories** in `VC.LEGENDARY_ACCESSORIES` suffers from live implementation bugs where actual combat code diverges from its authored design:

1.1 `LEG_ACC_SAMSARA_KARMA_RING` (แหวนหยกสังสารวัฏพลิกชะตา — `RING`, `lines 599–629, 905–917`)

- **Authored Tooltip (`line 622`)**: *"เมื่อโจมตี**หรือถูกตี** จะ**แปรเปลี่ยนสถานะร้าย *1* อย่างบนร่าง**ให้เป็นปราณน้ำค้างฟื้นฟู ดูดซับเลือด/ปราณ *6%* และสร้างความเสียหายวิบากกรรมเพิ่ม *+16%*"*
- **Three Live Bugs**:
  1. **Zero Debuff Cleanse (`lines 905–917`)**: Despite promising *"แปรเปลี่ยนสถานะร้าย *1* อย่างบนร่าง"*, the handler **never calls `ST.clearStatus`** (`0` debuffs removed or transmuted!).
  2. **Never Fires When Struck (`"หรือถูกตี"`)**: `VC.runWeaponOnStruckProcs` (`lines 918–940`) completely omits `EFX_LEG_SAMSARA_RING_REVERSAL`! If a wearer is stunned, silenced, or tanking hits without attacking, the Crisis Ring (`"ยามผู้สวมใส่ตกอยู่ในวิกฤต"`) **never activates**!
  3. **Bypasses `healBlock` (`line 908`)**: Calls raw `COMBAT.setHp(...)` instead of `COMBAT.healUnit`.

1.2 `LEG_ACC_BODHI_STARLIGHT_BEADS` (ประคำโพธิ์ดาราสว่างกระจ่างจิต — `NECKLACE`, `lines 630–659, 918–928`)

- **Authored Tooltip (`line 652`)**: *"สลายจิตมาร *(-6)* ฟื้นฟูพลังชีวิต *8%* **แก่ผู้สวมใส่และสหาย** พร้อมมอบบัฟบัวทิพย์กระจ่างใจและโคมวิญญาณคุ้มชีพ"*
- **Three Live Bugs**:
  1. **Zero Ally Healing (`lines 922–923`)**: Only heals `attacker` (`0` HP restored and `0` `nirvanaGrace` / `clarityLotus` shared with nearby injured allies, violating *"แก่ผู้สวมใส่และสหาย"*!).
  2. **Healers Casting Support/Heal Moves Can NEVER Trigger It (`line 955`)!**: `VC.runWeaponProcs` is hooked **exclusively** inside `COMBAT.strike` (`if (res && res.hit)`). When a `SAGE_SUPPORT` physician or

`LOTUS_MERCY_PURIFIER` monk casts non-damaging `heal`, `harmonize`, `support`, or `guard` skills on teammates, `COMBAT.strike` never runs—so **pure healers get `0` procs from the ultimate Healer Relic**!

3. **Missing `'BUDDHA_DHARMA'` in `compatiblePath` (`line 648`)**: Lists `['HEART_SOUL', 'SAGE_SUPPORT', 'GOD', 'BODY', 'ANY']`—omitting `'BUDDHA_DHARMA'` and all 6 Buddhist subpaths!

1.3 `LEG_ACC_TAICHI_HEAVEN_SEAL` (ตราพยัคฆ์หยกไท่จี๋บัญชาสวรรค์ — `CHARM`, `lines 660–688, 929–943`)

- **Two Live Bugs**:
  1. **Self-Only Command Tally (`lines 935–936`)**: Lore states "เปล่งบารมีบัญชาฟ้าดินและประสานพลังรบของพวกพ้องเป็นหนึ่งเดียว" (an Imperial Commander's Tiger Tally 虎符!), yet `tacticalCommand` and `meridianUnblock` are applied **only to `attacker`**, giving `0` command aura to allied vanguard/constructs within command range (`<= 4.5` tiles), and never triggers on `command` archetype casts (`useSkill`)!
  2. **80% Dead `preferredSubs` (`line 679`)**: `['GRAND_STRATEGIST', 'IMPERIAL_PEACE_SOVEREIGN', 'IRON_LEGION_MARSHAL', 'GRAND_FORMATION_ARCHITECT', 'QIMEN_TALISMAN_PATRON']` — **4 out of 5 subpath IDs were renamed/deleted** (`BATTLEFIELD_STRATEGIST`, `IMPERIAL_SEAL_CHANCELLOR`, `GRAND_LOGISTICS_COMMANDER`, `EARTH_VEIN_WARDEN` get `+0` suitability bonus)!

1.4 `LEG_ACC_DRAGON_PHOENIX_BANGLE` (กำไลหยกมังกรหงส์คู่สวรรค์ — `BRACELET`, `lines 689–717, 944–955`)

- Tooltip promises "เมื่อ**ร่ายวิชา**หรือโจมตี", yet non-damaging `support`/`heal`/`meditate`/`stance` skill casts in `COMBAT.useSkill` never invoke `VC.runWeaponProcs`; `COMBAT.setHp` bypasses `healBlock`; and `preferredSubs` (`line 707`) contains nonexistent `'PILL_FLAME_SOVEREIGN'` & `'SYMBIOTIC_FANG_FUSION'` (should be `'PILL_FURNACE_IMMORTAL'` / `'NINE_REVOLUTIONS_ALCHEMIST'` & `'SYMBIOTIC_BEAST_WARRIOR'`, plus Dual-Cultivation `RED_THREAD` spouses!).

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2. `[ACC-TAG-02]` Subtype Exclusion Bug: `'PENDANT'`, `'ORNAMENT'` & `'SHOES'` Lose Subpath Accessory Bonuses & Set Matching (`js/martial.js:495, 329`)

- **Exact Code (`js/martial.js:495`)**:

```javascript
const xMatch = !!(xType && ['RING', 'NECKLACE', 'BRACELET',
'CHARM'].includes(xType));
```

- **Impact**:
    1. Any accessory with `subType: 'PENDANT'`, `'ORNAMENT'`, or `'SHOES'` evaluates `xMatch === false` in `M.evaluateSubEquipmentBonus`, losing **`+3..5%` primary stats and flat `crit`/`pen`/`cast`/`acc`**!
    2. In `M.itemMatchesResonanceSet` (`martial.js:329`), `rs.slots.ACCESSORY` only lists `['RING', 'NECKLACE', 'BRACELET', 'CHARM']`, locking `'PENDANT'`, `'ORNAMENT'`, and `'SHOES'` out of Resonance Set completion!
    3. **Give `'SHOES'` Its Own Tactical Bonus in `evaluateSubEquipmentBonus`**: Instead of collapsing Boots into a generic `CHARM` stat block, map `'PENDANT' -> 'NECKLACE'`, `'ORNAMENT' -> 'CHARM'`, and recognize **`'SHOES'`** as a 5th valid accessory sub-archetype granting **`{ p: { eva: +0.05 }, f: { move: +6 } }`** (matching `RING`/`NECKLACE`/`BRACELET`/`CHARM` budget while rewarding scouts, couriers, and kiting archers)!

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3. `[ACC-SUIT-03]` Zero Role/Subpath Fit Differentiation for Accessories in `M.gearSuitability` (`js/martial.js:531–545`)

- Look at `js/martial.js:531–537`: `WEAPON` computes `fit` (`0.72..1.70`) and `ARMOR` computes `fit` (`0.85..1.25`), **whereas `ACCESSORY` leaves `fit = 1.0` flat across all 286 subpaths**!
- As a result:
    - Pure `SHADOW_SAMURAI` / `BODY` physical fighters (`matk` useless!) evaluate a `CHARM` (`matk +5%, mdef +4%`) almost identically to a `RING` (`atk +3%, crit +3, pen +4`) or `BRACELET` (`atk +4%, def +4%`).
    - Scouts (`scout`), Herbalists, and `WANDERING_DAO` disciples show **zero** preference for Cloud-Step `SHOES` (`ITEM190_EQ_017`) or Qiankun Supply Pouches (`ITEM190_EQ_018`).
- **Fix**: Add role-aware `ACCESSORY` `fit` weighting in `M.gearSuitability`:
    - `RING`: `+0.14` `fit` for crit/burst DPS (`role.atk >= 1.15` or assassin/archer subpaths).
    - `NECKLACE`: `+0.14` `fit` for `TANK` / `HEALER` / monastic subpaths (`role.def + role.mdef >= 2.35`).

- `BRACELET`: `+0.12 fit` for hybrid/fast-cast bruisers.
- `CHARM`: `+0.14 fit` for `CASTER` / `ARRAY` / `TALISMAN` (`role.matk >= 1.20`).
- `SHOES`: `+0.18 fit` for `WANDERING_DAO`, `NINPO_SHADOW`, `BOW` kiters, and `scout`/`wildwarden` officers!

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4. `[ACC-AUTH-04]` Authored Personal Wearables (`ITEM190_EQ_014..018` in `data_items.js`): Unwired World Effects & Raw Hanzi!

Inspecting `data_items.js` (`ITEM190_EQ_014..018`) against `economy.js`, `cultivation.js`, `combat.js`, and `expedition.js` reveals **five broken or incomplete wearable loops**:

1. **`ITEM190_EQ_018` (`"ถุง乾坤จัดสรรเสบียงเดินทาง"` — Qiankun Spatial Supply Pouch)**:
   - **Bug A (Untranslated Hanzi)**: `nameTh` contains raw Chinese `乾坤` inside Thai text! Rename to `"ถุงเฉียนคุนจัดสรรเสบียงเดินทาง"` (also fix `REC190_EQ_018`).
   - **Bug B (`EFX_SUPPLY_EFFICIENCY` 100% Dead in `expedition.js`!)**: `supplySavingRatio: 0.22` is never read in `EXPED.depart` or `EXPED.createParty` (`economy.js:995` writes `npc.ext.supplyEfficiencyPct`, which **zero** files in the entire codebase read—and only when consumed, not when equipped!).
   - **Fix**: In `EXPED.createParty` / `EXPED.depart`, inspect party members' usable equipped `ACCESSORY` for `EFX_SUPPLY_EFFICIENCY` / `EFX_CARGO_PRESERVATION`, reducing expedition `meal` ration cost by `clamp(sum, 0, 0.30)` (min `1` meal) and increasing `carryCapacity` by `+25%`!

2. **`ITEM190_EQ_014` (หยกพกสะสมปราณเก้าวัฏจักร — `EFX_QI_RESERVOIR`, `35` catalog items!)**:
   - `equip_repro_result.log` shows **`EFX_QI_RESERVOIR` is the #4 most common effect in the entire equipment catalog (`35` items!)**, defining `{ reservoirCapacity: 380, leakagePerDay: 2 }`.
   - Yet when worn in `equip.ACCESSORY`, there is **no Qi Reservoir bank** (`acc._storedQi`) charged from surplus meditation Qi (`A.cultivate`) and automatically tapped in combat when `n.h.state.qi < 25% qiMax`! Implementing this lightweight 6-line reservoir makes all **35 `EFX_QI_RESERVOIR` accessories** genuinely unique!

3. **`ITEM190_EQ_016` (ประคำโพธิ์ขาวสงบจิตมาร — `EFX_CALM_MIND`)**:
    - Defines `{ heartDemonReduction: 24, soulDefenseBonus: 40 }`. Currently `EFX_CALM_MIND` only fires in `pillInfo` / `executeItemEffectById` (consumables!). When worn as an accessory, wire `ST.gearStats` (`+mdef`) and `A.cultivate.tick` / `CULT.attemptBreakthrough` (`-20%` `heartDemon` buildup & `+0.04` mental-state breakthrough stability)!
4. **`ITEM190_EQ_017` (รองเท้าเมฆาคล้อยพริ้วไหว — `EFX_SIDESTEP_EVASION`, `SHOES`)**:
    - Stamp `subType: 'SHOES'`, grant `+8%` map `walkSpeed` in `M.derived` (`martial.js:183`) when usable `SHOES` are equipped, and trigger `evasionChanceBonus` first-step dodge in `COMBAT`!
5. **`ITEM190_EQ_015` (จี้เนตรพยัคฆ์สดับจิตสังหาร — `EFX_INTENT_SENSE`, `NECKLACE`)**:
    - Stamp `subType: 'NECKLACE'`, grant `firstTurnEvasionBonus: +0.22` during Round 1 (`f.t - f.t0 <= 4`) in `COMBAT.strike` and `-20%` ambush surprise rate in `explore.js` / `expedition.js`!

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5. Implementation & Verification Checklist for GPT / Gemini (`05B`)

- ☐ **Fix All 4 Legendary Accessories (`js/visitors_clues.js`)**:
    - `SAMSARA_KARMA_RING`: Cleanse 1 hostile status on proc, trigger in `runWeaponOnStruckProcs` when hit, and route heal through `COMBAT.healUnit`.
    - `BODHI_STARLIGHT_BEADS`: Heal allies within `4.0` tiles (`"และสหาย"`), add `'BUDDHA_DHARMA'` to `compatiblePath`, and hook `VC.runSupportRelicProcs` into `COMBAT.useSkill` so non-damaging `heal`/`support`/`command` casts trigger support relics!
    - `TAICHI_HEAVEN_SEAL`: Broadcast `tacticalCommand` to nearby allies/constructs and replace the 4 stale `preferredSubs` IDs.
    - `DRAGON_PHOENIX_BANGLE`: Trigger on non-strike skill casts, respect `healBlock`, and fix stale `preferredSubs`.
- ☐ **Support `'SHOES'`, `'PENDANT'` & `'ORNAMENT'` (`js/martial.js`)**: Normalize `PENDANT -> NECKLACE`, `ORNAMENT -> CHARM`, add `'SHOES'` (`+5% eva, +6 move`) to `evaluateSubEquipmentBonus` & Resonance Set matching, and add role-aware `ACCESSORY` `fit` scoring in `M.gearSuitability`.
- ☐ **Wire Authored Personal Wearables (`ITEM190_EQ_014..018`)**: Fix raw Hanzi 乾坤 $\to$ เฉียนคุน, wire `EFX_SUPPLY_EFFICIENCY` into `expedition.js` rations/capacity,

wire `EFX_QI_RESERVOIR` (`35` items!) meditation-to-combat Qi bank, and wire `EFX_CALM_MIND`, `EFX_SIDESTEP_EVASION` (`walkSpeed`), and `EFX_INTENT_SENSE` Round-1 ambush evasion.